

Adapting the D - ^3He Levitated Dipole Rocket to p - ^{11}B for an Aneutronic Power Plant with Reduced Astronaut Shielding

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Abstract

The **Deuterium–Helium-3 (D - ^3He) Levitated Dipole Rocket (LDR)** is a systems-level *conceptual* architecture for crewed interplanetary propulsion: a high- β magnetospheric core, open-field magnetic nozzle, and ISRU water propellant dilution [1–4]. That baseline—including the mid-2026 finding that the LDR remains a **paper architecture** with **no commercial or dedicated government program**, while adjacent **OpenStar LDC** hardware targets **terrestrial deuterium–tritium (DT)** electricity—is surveyed in a companion manuscript at [catskillsresearch/ldc_fusion_github](https://github.com/catskillsresearch/ldc_fusion_github) (not yet deposited on arXiv or Zenodo) [5].

This paper asks a narrower systems question: **can that same dipole rocket framework be adapted—largely directly—to the proton–boron-11 (p - ^{11}B) fuel cycle**, yielding an **aneutronic** spacecraft power plant whose **crew radiation shield can be deleted or drastically thinned** relative to D - ^3He ? Among current p - ^{11}B concepts catalogued in the companion survey on Zenodo [6], the most apt graft is the **open-field, ash-first** line epitomized by Princeton / Pale Blue **CHARM** (differential confinement, alpha channeling, prompt helium removal) [6–9], with supporting high- β / beam-driven kinetics lessons from **TAE’s FRC** path [6,10]. Pulsed laser and pinch plants do not map cleanly onto a continuously levitated dipole rocket.

We conclude that the LDR’s **topology can be retained** (levitated ring, turbulent inward pinch, open outer field lines, water-diluted magnetic nozzle), while **fuel, kinetics, and ash handling must be retuned** for Rider / Bremsstrahlung limits and three-alpha ash. On shielding: the **thick neutron shadow shield of the D - ^3He LDR can be largely removed**; a **thinner high- Z X-ray / Bremsstrahlung shadow** remains, and galactic cosmic-ray protection for the habitat is unchanged.

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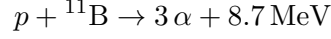
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1 Introduction

Crewed fusion spacecraft are mass-limited by the rocket equation and by **radiation-shield dry mass**. The $D\text{-}^3\text{He}$ LDR was designed to ease both: high specific impulse with tunable thrust via water dilution, and a **directional neutron shadow shield** sized for the residual $\sim 1\text{--}5\%$ neutron power fraction of a ^3He -rich $D\text{-}^3\text{He}$ plasma [1,2,5].

$p\text{-}^{11}\text{B}$ raises the cleanliness bar further:



Primary products are **fully charged**. Residual neutrons from secondaries such as $^{11}\text{B}(\alpha, n)^{14}\text{N}$ and $^{11}\text{B}(p, n)^{11}\text{C}$ typically carry $\lesssim 0.2\%$ of fusion energy—inside common “aneutronic” definitions ($\lesssim 1\%$), versus several percent for $D\text{-}^3\text{He}$ side channels [6]. That is the engineering invitation: **delete most of the neutron moderator mass** that dominates crewed $D\text{-}^3\text{He}$ (and D-T) shield stacks.

The catch is well known: a steady, Maxwellian, well-mixed $p\text{-}^{11}\text{B}$ plasma is constrained by the **Rider limit**—Bremsstrahlung from hot electrons can outrun fusion power [6,11]. Modern programs therefore do **not** pitch “just heat a soup.” They pitch **nonthermal or differentially confined** kinetics, **prompt alpha removal**, and (where possible) **high- β** suppression of synchrotron radiation [6]. The levitated dipole already supplies high $\beta \sim 1$ and **open outer field lines**—exactly the geometric features those programs demand.

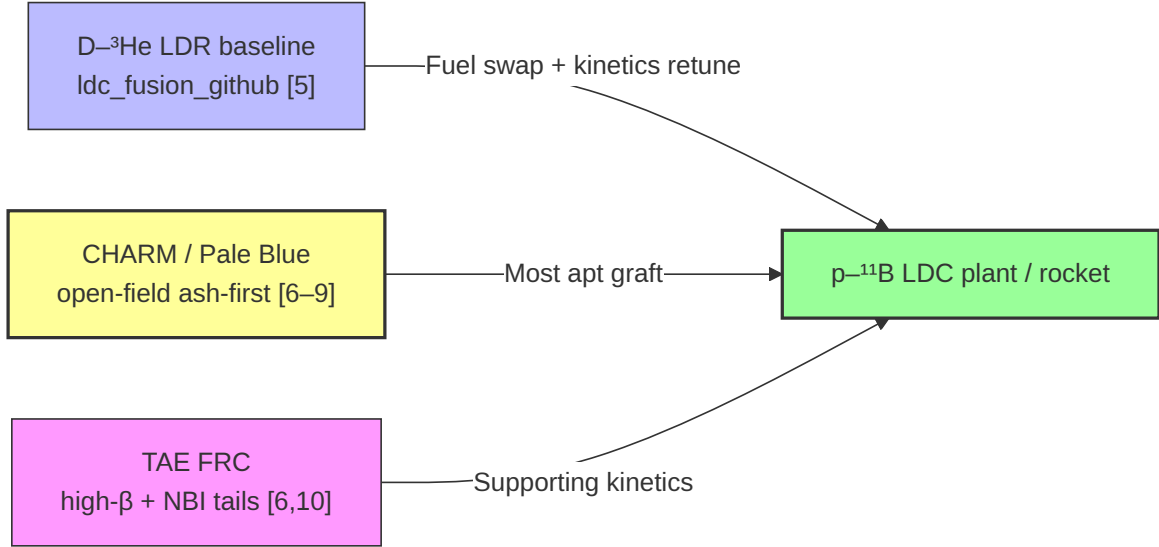


Figure 1 Introduction.

2 Baseline: What the $D\text{-}^3\text{He}$ LDR Already Provides

We retain the architecture of [1–5] without reopening its 1987–1994 historical lineage. Per [5] §7, that architecture is a **conceptual** LDR: transferable physics and (via adjacent terrestrial LDC work) magnet engineering—not an active rocket product. The reusable blocks are:

1. **Levitated superconducting dipole** with magnetospheric confinement and a **turbulent inward pinch** that peaks density toward the ring ($n \propto r^{-4}$ class profiles) at local $\beta \sim 1$ [3,5,12].

2. **Open outer field lines** that divert heat and particles into an **axial magnetic nozzle**—already a rocket exhaust, and already a natural ash dump [1,2,4,5].
3. **ISRU water injection** into the nozzle throat for thrust / I_{sp} modulation ($I_{sp} \sim 5,000\text{--}50,000$ s class in the baseline survey) [4,5].
4. **Modular ring docking** for in-flight cryogenic service and replacement—historical LDR maintainability plus REBCO / flux-pump lessons from adjacent **terrestrial DT** LDC hardware [5,14,15].
5. A **shadow-shield cone** protecting the habitat: tungsten (X-rays) + borated hydrogenous moderator (neutrons) [2,5].

Items 1–4 transfer to $p\text{--}^{11}\text{B}$ with modest retuning. Item 5 is where mass can be cut.

3 Choosing Among $p\text{--}^{11}\text{B}$ Concepts: Why Open-Field / Ash-First Fits

The companion catalog [6] sorts commercial and theory paths by time (continuous vs pulsed), confinement family, fuel end-state, and kinetics. For a **continuously levitated dipole rocket**, most leaves are a poor fit (Table 1):

Table 1 Fit of proton-boron concepts to the LDR / LDC framework

Path (from [6])	Fit to LDR / LDC	Why
Pale Blue / CHARM (rotating multi-chamber open-field mirror)	Best	Continuous; open-field; designed around species separation and He ash strain ; alpha channeling reduces required $n\tau_E$ [6–9]
TAE FRC (beam-driven, high- β)	Strong supporting lessons	High- β core; open ends for ash / DEC; NBI non-Maxwellian proton tails [6,10]—kinetics portable even if the bottle is a dipole, not an FRC
ENN spherical torus	Weak topology match	Closed torus; no natural rocket nozzle; different maintenance geometry [6]
HB11 / laser ICF, LPP DPF	Poor	Pulsed drivers; no levitated steady magnet; incompatible with LDR operations concept [6]
Orbitron / NVD–IEC	Poor as the primary plant	Compact electrostatic / MEC class; not a magnetospheric rocket core [6]

Design choice. Treat the adapted vehicle as a $p\text{--}^{11}\text{B}$ **levitated-dipole plant with CHARM-inspired differential confinement and ash exhaust on the existing open field lines**, optionally assisted by **TAE-like neutral-beam or RF-driven proton tails**. Do

not replace the dipole with an FRC or a laser chamber; borrow their **physics modules**, not their vacuum vessels.

4 Direct Adaptation: Fuel, Kinetics, and Ash on the Same Dipole

4.1 Fuel cycle (direct swap)

Replace D - ^3He with a **proton-rich** p - ^{11}B mix ($n_p/n_B > 1$). Proton richness limits Bremsstrahlung relative to boron-heavy stoichiometry and is standard in low-density magnetic p - ^{11}B modeling [6]. Fuel logistics improve: ^{11}B and hydrogen are Earth-abundant; scarce lunar ^3He is no longer mission-critical [5,6].

4.2 Why high- β still matters (synchrotron)

p - ^{11}B wants ion energies well above the D - ^3He window (reactivity peak near ~ 150 – 300 keV class ions in magnetic concepts) [6]. Synchrotron losses scale unfavorably with B at fixed pressure; the LDR's $\beta \sim 1$ operating point **minimizes the required field** for a given plasma pressure—the same argument used for D - ^3He [1–3,5], now even more load-bearing.

4.3 Rider / Bremsstrahlung: required kinetic tweak (not a topology change)

A thermalized $T_i = T_e$ soup is not the intended point of operation [6,11]. The adapted plant should target one or both of:

- **Hot-ion / cooler-electron window** (near-equilibrium soft window with constrained T_i/T_e), as in recent 0D Fokker–Planck reassessments [6]; and/or
- **Nonthermal proton population** near the reactivity peak via NBI or wave drive (TAE-style tails; Fisch / CHARM alpha-channeling into fast protons) [6–10].

The dipole does not invent these loopholes; it **hosts** them. Electron temperature control and beam / RF hardware are the new plant systems—not a new bottle.

4.4 Three-alpha ash: use the open field lines (CHARM lesson on LDR geometry)

Three ^4He nuclei per reaction make **ash poisoning** existential: if helium lingers for an energy-confinement time, engineering breakeven can fail even with perfect τ_E [6–8]. CHARM's answer is architectural—**separate fusion, boron, and ash / heat-exchange regions** with one-way RF / ponderomotive walls [6,9].

On the LDR, the natural analogue is:

- **Prompt alpha exhaust** along open outer lines toward the nozzle / end converters (FRC end-cell intuition [10], mapped onto the LDR divertor already present in [1,2,5]).
- **Optional CHARM-like barriers** to keep cold boron and hot protons from forming a single Rider soup, and to **strain He** into a dedicated heat-exchange / direct-conversion zone [7–9].
- **Alpha channeling** (waves preferentially heating protons rather than electrons) as a power-flow retune that can cut required confinement product by factors of a few to ~ 7 in published 0D models [7,8]—again, a wave system on the existing plasma, not a new magnet.

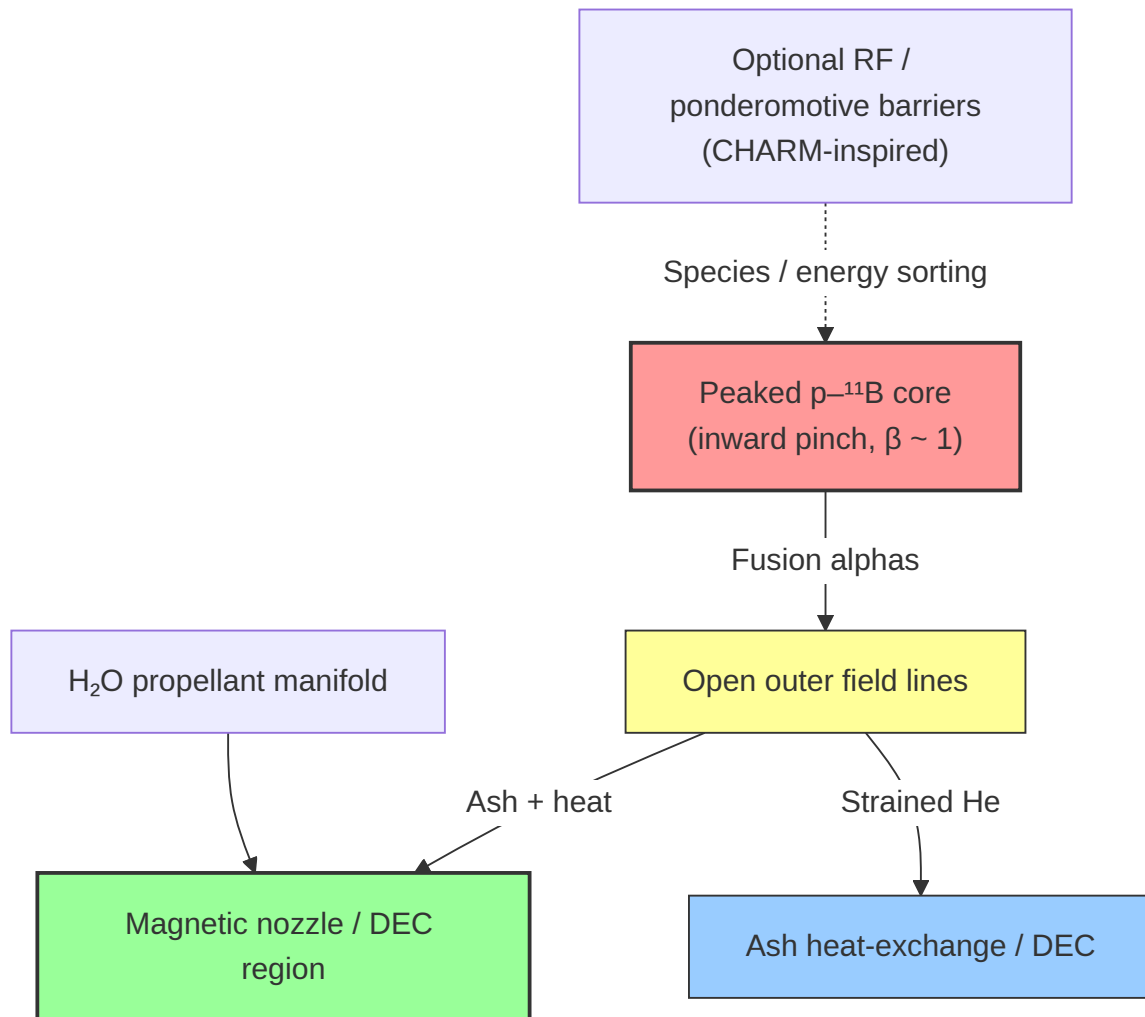


Figure 2 Three-alpha ash: use the open field lines (CHARM lesson on LDR geometry).

Verdict on “direct vs tweak.” Geometry: **direct**. Fuel: **direct swap**. Kinetics and ash: **necessary tweaks**, best informed by CHARM + TAE rather than by closed-torus ST roadmaps.

4.5 Propulsion vs shipboard power

The baseline LDR is a **rocket**. A $p\text{-}^{11}\text{B}$ adaptation naturally supports a dual role:

- **Thrust mode:** fusion alphas (and any unburned ions) enter the magnetic nozzle; **ISRU water** still provides mass flow for kilonewton-class thrust and I_{sp} modulation [4,5]. Exhaust remains fully ionized and solar-wind-dispersible [5].
- **Power-plant mode:** divert a larger fraction of charged products into **direct energy conversion** (traveling-wave / inductive converters on expanding field lines—TAE end-cell class ideas [10]) to feed habitat, cryogenics, NBI/RF, and avionics **without** a heavy neutron-activated blanket.

Crewed missions likely need both: continuous shipboard electricity and episodic high-thrust burns.

5 Shielding: What Can Be Deleted, What Must Remain

5.1 Neutron shield — largely delete

In the $D\text{-}^3\text{He}$ LDR, secondary D–D / D–T neutrons force a **hydrogenous moderator + boron capture** layer behind tungsten [2,5]. For $p\text{-}^{11}\text{B}$:

- Neutron **energy fraction** drops from percent-level to $\sim 0.1\text{--}0.2\%$ class [6].
- There is **no tritium breeding blanket** and no intentional deuterium inventory (deuterium contamination must be controlled as an impurity, not a fuel) [6].
- Open-field divertor geometry already keeps the vacuum vessel from becoming a highly activated tokamak-like wall [5].

Systems recommendation: remove the thick borated-water / polyethylene stack from the habitat shadow cone. Retain at most a **thin residual-neutron liner** (or none, if dose calculations for the chosen mix stay within mission limits with geometry and distance alone). This is the primary dry-mass win for astronauts.

5.2 X-ray / Bremsstrahlung shield — keep, possibly thin differently

Hard X-rays remain an engineering radiation load even when neutrons are negligible [6]. A high- Z **tungsten (or equivalent) shadow** facing the core is still warranted for habitat dose control—especially if T_e or Z_{eff} excursions raise Bremsstrahlung. Thickness should be set by X-ray transport, **not** by 14 MeV neutron attenuation; expect a **lighter** cone than the $D\text{-}^3\text{He}$ dual-layer shield, not a zero-shield utopia.

5.3 What shielding does *not* go away

Interplanetary **galactic cosmic rays (GCR)** and solar particle events still require habitat mass (water walls, regolith, storm shelters). That mass is **mission physics**, not reactor neutronics. The $p\text{-}^{11}\text{B}$ adaptation removes the **reactor-neutron** penalty; it does not cancel cosmic-ray engineering.

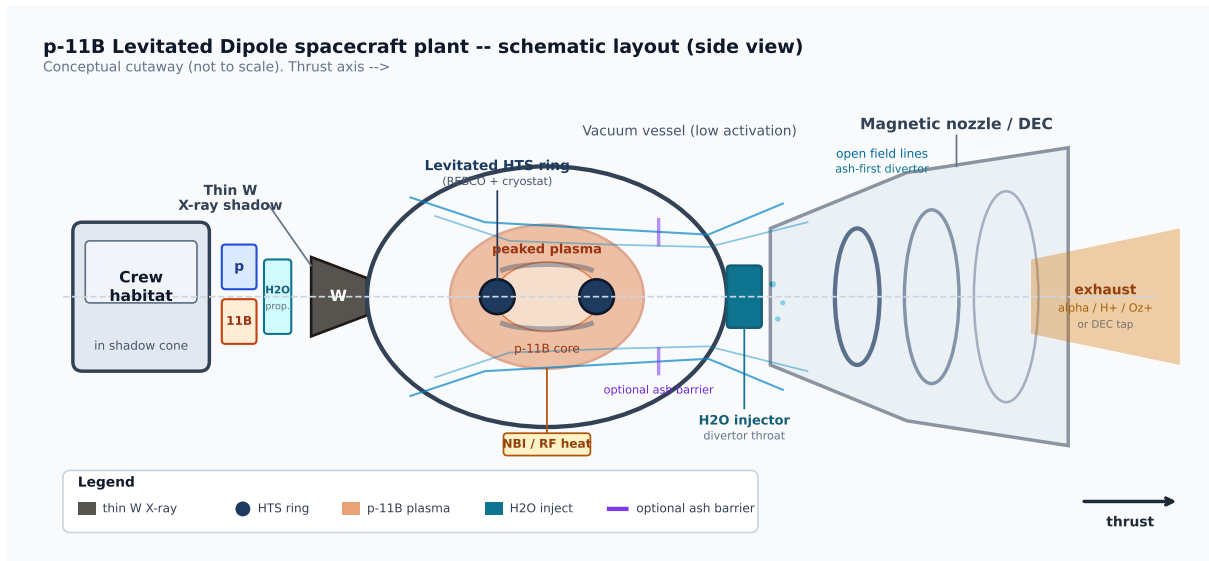


Figure 3 Schematic side view of the $p\text{-}^{11}\text{B}$ Levitated Dipole spacecraft plant. Crew-side (shadow cone): habitat, compact $p\text{-}^{11}\text{B}$ fuel tanks, and a separate H_2O propellant tank. Shadow shield: thin tungsten X-ray absorber only (no thick neutron stack). Aft: levitated HTS ring and peaked $p\text{-}^{11}\text{B}$ plasma, optional ash barrier on open field lines, NBI/RF heating, H_2O injector at the divertor throat, magnetic nozzle / DEC.

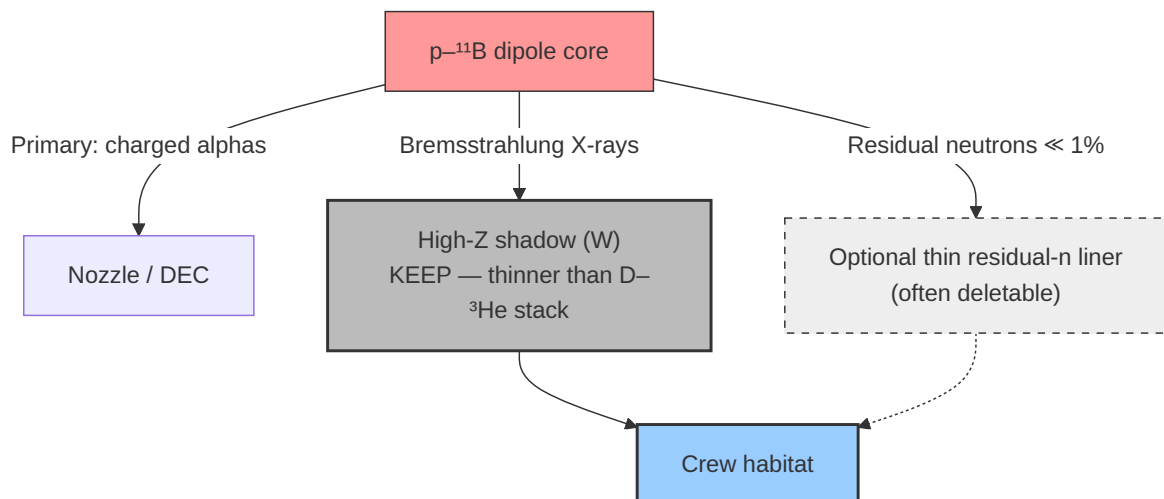


Figure 4 X-ray / Bremsstrahlung shield — keep, possibly thin differently.

5.4 Maintainability upside

With neutrons nearly gone, post-shutdown activation of the chamber and external hardware drops further below the already favorable D - ^3He open-field case [5]. EVA access after scram improves; modular ring swap remains the preferred HTS maintenance path [5].

6 Mass and Mission Implications (Qualitative)

Relative to the D - ^3He LDR survey [5] (Table 2):

Table 2 Subsystem mass and role comparison

Subsystem	D - ^3He LDR	p - ^{11}B adaptation
Levitated ring + cryostat	Required	Required (REBCO / flux-pump class as in adjacent terrestrial DT LDC hardware [5,14,15])
Fuel	Scarce ^3He	Abundant p , ^{11}B
Neutron moderator shield	Required (1–5% P_n)	Delete / minimal ($\lesssim 0.2\%$ P_n)
High- Z X-ray shadow	Required	Required, thinner budget
NBI / RF / wave systems	Auxiliary	First-class (Rider + ash)
Water propellant / nozzle	Core to thrust	Retained for rocket mode; DEC emphasized for power mode

The dry-mass story is therefore: **trade neutron shield mass for beam/RF and ash-handling hardware**, with a favorable exchange rate if CHARM-like ash control closes, because shield mass scales with solid angle and mission dose while converters scale with power.

7 Risks and Open Problems

This adaptation is a **systems sketch**, not a closed Lawson design.

1. **Net Q on p - ^{11}B** is unproven in any magnetic plant [6]; the dipole inherits that field-wide gap.
2. **Integrated self-consistency** of high- β dipole profiles with nonthermal protons, cooler electrons, and three-alpha exhaust needs 0D→transport modeling (CHARM’s in-silico-first culture is the right diligence style [6,9]).
3. **RF barrier / species separation** hardware on a levitated dipole is conceptual—ported from CHARM patents and theory [9], not demonstrated on LDX-class devices.
4. **Bremsstrahlung still heats walls**; X-ray shield and first-wall thermal design remain critical even after neutrons vanish [6].

5. Per [5] §7: **no company and no dedicated academic/government program** is building a $D\text{-}^3\text{He}$ LDR; the only commercial LDC firm (**OpenStar**) targets **terrestrial DT** electricity [14,15]. The same gap applies *a fortiori* to a $p\text{-}^{11}\text{B}$ LDC rocket—fuel-cycle R&D must be imported from the $p\text{-}^{11}\text{B}$ community [6].

8 Compare and Contrast: $p\text{-}^{11}\text{B}$ versus the Original $D\text{-}^3\text{He}$ LDR

Sections 5–7 argue subsystem-by-subsystem. This section states the trade explicitly: **feasibility**, **cost / mass economics**, and **mission improvements** of the $p\text{-}^{11}\text{B}$ adaptation relative to the original $D\text{-}^3\text{He}$ LDR [1–5].

8.1 Feasibility

Table 3 Feasibility compare-and-contrast

Dimension	Original $D\text{-}^3\text{He}$ LDR	$p\text{-}^{11}\text{B}$ adaptation	Edge
Primary reaction cleanliness	Charged products, but D–D / D–T side neutrons at 1–5% of fusion power [2,5]	Primary channel fully charged; residual neutrons typically $\lesssim 0.2\%$ [6]	$p\text{-}^{11}\text{B}$
Plasma physics difficulty	Hard, but historically sized for dipole rockets at $T_i \sim 50\text{--}100\text{ keV}$ [1–3]	Harder: Rider / Bremsstrahlung bound on thermal soups; needs hot-ion or nonthermal kinetics and prompt three-alpha ash removal [6,11]	$D\text{-}^3\text{He}$ (nearer-term physics)
Bottle / topology readiness	Designed <i>for</i> the levitated dipole + magnetic nozzle [1,2,4]	Same bottle retained; open lines already match CHARM / FRC ash-exhaust intuition [6–10]	Tie (geometry already fits)
Fuel availability	^3He scarce on Earth; lunar / ISRU sourcing assumed [5]	Protons and ^{11}B Earth-abundant; no ^3He logistics [6]	$p\text{-}^{11}\text{B}$
Demonstrated plant Q	Neither fuel has a flown or utility-closed dipole plant; $D\text{-}^3\text{He}$ has a longer peer-reviewed space-propulsion paper trail [1–4]	No magnetic $p\text{-}^{11}\text{B}$ plant has closed engineering Q yet [6]	$D\text{-}^3\text{He}$ (literature maturity)
Crew maintainability after scram	Already favorable (open-field, low activation vs tokamaks) [5]	Better still: neutron activation nearly eliminated [6]	$p\text{-}^{11}\text{B}$

Feasibility bottom line. The original design is the **more credible near-term physics path** (easier ignition metrics, decades of LDR-specific analysis). The $p\text{-}^{11}\text{B}$ adaptation is the **more credible long-term cleanliness path**, but it **raises** the plasma-control bar (Rider, ash) even while it **lowers** the neutronics and fuel-logistics bars. It is not a free upgrade; it is a deliberate bet that kinetics and ash modules can be closed on a bottle that already looks right.

8.2 Cost and mass economics

Dollar costs for either concept are not bankable today; what *is* comparable is **where mass and recurring cost sit** (Table 4):

Table 4 Cost and mass-driver comparison

Cost / mass driver	Original $D\text{-}^3\text{He}$ LDR	$p\text{-}^{11}\text{B}$ adaptation
Reactor neutron shield	Thick borated hydrogenous layer in the habitat shadow cone [2,5] — recurring structural mass every mission	Largely deleted ; optional thin residual liner only [§5]
X-ray shadow	Tungsten (or equivalent) required	Still required; budget set by Bremsstrahlung, not 14 MeV n — typically lighter than the dual-layer $D\text{-}^3\text{He}$ stack
Fuel recurring cost	^3He purchase or lunar mining dominates fuel economics [5]	Commodity boron + hydrogen / water — orders of magnitude cheaper per joule of fuel inventory
Auxiliary plant (NBI, RF, wave, DEC)	Present but secondary to fusion + nozzle	First-class capital : beams/waves and ash converters replace neutron shield as the expensive add-on [§4, §6]
Activated-structure / spares	Low vs D-T tokamaks; still nonzero secondary- n activation [5]	Lower still — fewer activated spares and simpler EVA dose planning
ISRU water propellant	Core to thrust economics [4,5]	Unchanged for rocket mode

Cost bottom line. $p\text{-}^{11}\text{B}$ **cuts shield dry mass and fuel logistics cost** and **adds** beam/RF/ash-handling capital. The exchange is favorable for crewed vehicles when shield mass is a dominant fraction of reactor-related dry mass—the usual LDR motivation [2,5]—**if** the added kinetic systems do not erase that savings. GCR habitat mass is common to both and should not be booked as a $p\text{-}^{11}\text{B}$ win.

8.3 Improvements (what you actually gain)

Relative to the original LDR, a successful $p\text{-}^{11}\text{B}$ adaptation buys:

1. **Aneutronic-class crew dose from the reactor** — astronaut shadow shielding set by X-rays and operations, not by a continuous neutron flood [§5].
2. **Fuel independence from ^3He** — removes a strategic and economic single point of failure in the baseline propellant/fuel story [5,6].

3. **Cleaner dual-use power plant** — charged alphas favor **direct energy conversion** for shipboard electricity alongside the existing water-diluted nozzle for thrust [§4.5].
4. **Simpler radiological operations** — less activation, faster post-shutdown access, no tritium-breeding architecture [5,6].
5. **Alignment with the modern $p\text{-}^{11}\text{B}$ R&D ecosystem** — CHARM ash-first and TAE high- β / beam kinetics are active programs whose modules map onto the LDR geometry [6–10], whereas waiting on lunar ^3He does not.

What you **do not** automatically gain: easier plasma physics, proven net gain, or relief from cosmic-ray shielding.

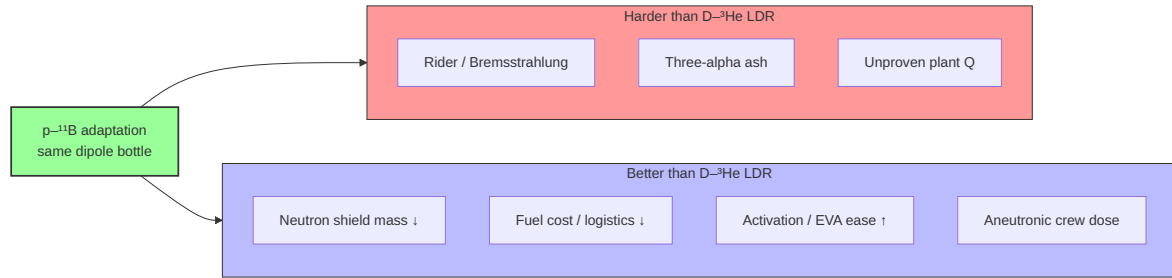


Figure 5 Improvements (what you actually gain).

8.4 Who is actually building what (aligned with [5] §7)

The baseline survey [5] §7 separates three scopes that must not be conflated: (i) the $D\text{-}^3\text{He}$ LDR paper architecture; (ii) **commercial LDC hardware for terrestrial electricity** (DT); (iii) **active academic/government LDR programs**. Mid-2026 status, matching [5] and public sources (Table 5):

Table 5 Commercial and program ownership of LDC / LDR theses

Thesis	Owner / program?	Notes
Levitated dipole bottle (commercial)	Yes — only OpenStar Technologies (Wellington, NZ)	Sole commercial LDC developer (Junior / Tahi); peer-reviewed terrestrial DT plant concepts—not space propulsion, not $D\text{-}^3\text{He}$ [5,14,15]

Thesis	Owner / program?	Notes
Original D-^3He LDR (dipole + ^3He + water nozzle rocket)	No company; no dedicated academic/gov program	1991–1994 conceptual line [1–4]; remains a paper architecture [5]. OpenStar shares transferable bottle tech (REBCO, flux pump, modular ring), not the LDR product thesis [5]. Helion / PFS pursue D - ^3He on FRC / PFRC , not LDC [6]. LDX / RT-1 were terrestrial confinement experiments, not rockets [5].
This paper’s p-^{11}B LDC adaptation	No	Imputed / paper-plant path. CHARM and TAE supply kinetics modules on <i>other</i> bottles [6–10]; nobody is burning p - ^{11}B in a levitated dipole.

Takeaway (same as [5] §7.6). One company owns commercial LDC hardware; **zero** companies (and no dedicated government line item) own the D - ^3He LDR; **zero** own a p - ^{11}B LDC rocket/plant.

8.5 Diligence scorecard and Plant Odds Score (adapted from [6] §12)

We reuse the zeroth-order gates and **Plant Odds Score (POS $\star$)** of the companion p - ^{11}B survey [6] (§1.7 diligence gates; §8 scorecard; §12.1 POS $\star$), with three spacecraft-relevant adaptations:

1. **Success definition** — first usable **crewed spacecraft fusion plant** (engineering $Q \gtrsim 1$ with thrust and/or shipboard power), not grid electricity.
2. **Gate reading** — **L** = closed power/thrust plant gain (not MWe busbar); **M** includes nozzle, DEC, and **astronaut reactor-shield mass**; **F** includes mission fuel logistics (^3He scarcity vs abundant p / ^{11}B); **T/H** score *ownership of this fuel+architecture*, not a related bottle with a different fuel.
3. κ — both rows are **imputed / hypothetical plants** ($\kappa = 0.50$ in [6] §12), because neither has a company or dedicated government program executing the scored end-state ([5] §7). OpenStar’s **terrestrial DT** LDC program is **not** folded into either row’s **T/H** (wrong fuel and wrong product thesis).

Legend (same as [6]): • = articulated with supporting analysis or hardware elsewhere in the lineage (score 2); $\odot$ = partial / roadmap / ported (1); $\circ$ = weak or absent (0).

Table 6 is the editorial mid-2026 scorecard with paths as columns (only two paths) and each gate labeled by letter and name.

Table 6 Diligence scorecard for LDR paths (gates as rows)

Gate	D - ^3He LDR [1–5]	p - ^{11}B LDC adaptation (this work)
Type	Imputed rocket plant	Imputed rocket / plant
C — Confinement class	LDC	LDC
F — Fuel & nuclear (+ logistics)	⊙	●
K — Kinetics / Rider	●	⊙
R — Radiation	⊙	⊙
A — Ash & impurities	⊙	⊙
L — Lawson / engineering Q	⊙	○
M — Materials, capture, crew shield	⊙	⊙
T — Technology-to-market	○	○
S — In-silico / digital twin	⊙	⊙
H — Hardware iteration	○	○
Notes	Paper architecture ([5] §7); ^3He logistics softens F ; OpenStar LDC H is terrestrial DT, not credited here	Abundant fuel (F); Rider/ash need CHARM/TAE grafts (K/A); no closed L ; no owner (T/H)

POS formula (unchanged from [6] §12.1), with ●= 2, ⊙= 1, ○= 0:

$$\text{POS} = \frac{2(K + R + A + L) + 1.5(T + H) + (F + M + S)}{28} \times 100, \quad \text{POS}^* = \kappa \times \text{POS}.$$

Table 7 Plant Odds Score (POS*) for the two LDR paths

Path	K, R, A, L	T, H	F, M, S	POS	κ	POS*
D - ^3He LDR	2 + 1 + 1 + 1 = 5	0 + 0 = 0	1 + 1 + 1 = 3	2 · 5 + 1.5 · 0 + 3 = 13 → 46	0.50	23
p - ^{11}B LDC adaptation	1 + 1 + 1 + 0 = 3	0 + 0 = 0	2 + 1 + 1 = 4	2 · 3 + 0 + 4 = 10 → 36	0.50	18

How to read the scores. Gaps below about 10 POS* points are noise in [6]; the **23** vs **18** spread in Table 7 is therefore a mild preference for the original LDR on the weighted physics+execution composite—driven by stronger **K/L** on published D - ^3He dipole-rocket sizing, not by commercial ownership (both **T/H** are empty). The p - ^{11}B column in Table 6 wins raw **F** and the shielding story inside **M**, but loses on **L** and pays the Rider/ash tax in **K/A**. Neither

path approaches company-owned $p\text{-}^{11}\text{B}$ leaders in [6] (e.g. TAE POS $\star \sim 79$) because those programs have real **T/H**; these LDC fuel theses do not.

If **OpenStar (or a new program)** later adopted either advanced fuel on a dipole rocket, the correct move would be to raise that row's **H** (and eventually **T**) and possibly $\kappa \rightarrow 1.0$ —not to invent a second LDC company. As of mid-2026 that adoption has not happened for $D\text{-}^3\text{He}$ rocket or $p\text{-}^{11}\text{B}$ [5].

8.6 Verdict

Prefer the **original $D\text{-}^3\text{He}$ LDR** if the priority is **nearest-term propulsion credibility** on published dipole-rocket physics (higher POS $\star$ here: **23 vs 18**). Prefer the **$p\text{-}^{11}\text{B}$ adaptation** if the priority is **crew shielding mass, fuel abundance, and aneutronic operations**—accepting a steeper plasma and ash-control climb on an otherwise unchanged levitated-dipole airframe. In both cases, treat the path as an **imputed plant**, consistent with [5] §7: **OpenStar is the only commercial LDC company, and it is building terrestrial DT plants—not either of these rocket fuel products.**

9 Conclusion

The $D\text{-}^3\text{He}$ Levitated Dipole Rocket [1–5] is already close to what $p\text{-}^{11}\text{B}$ magnetic concepts ask for: **high β , open field lines, and a nozzle-ready divertor**. Among contemporary $p\text{-}^{11}\text{B}$ designs [6], **CHARM's open-field, ash-first architecture** is the most apt conceptual graft, with **TAE-style beam/nonthermal proton kinetics** as a portable supporting module.

Direct: retain the levitated dipole, inward pinch, water-diluted magnetic nozzle, and modular ring. **Tweak:** proton-rich $p\text{-}^{11}\text{B}$ fuel, nonthermal / hot-ion kinetics against the Rider limit, and prompt helium exhaust (optionally with CHARM-like differential barriers). **Shielding:** delete the thick neutron moderator; keep a lighter high- Z X-ray shadow; leave cosmic-ray habitat mass alone.

As §8 lays out: $D\text{-}^3\text{He}$ wins on **near-term plasma feasibility** (POS $\star$ **23 vs 18** on the adapted [6] §12 scorecard); $p\text{-}^{11}\text{B}$ wins on **shield mass, fuel cost, and aneutronic crew dose**—if kinetics and ash modules close. Consistent with the corrected baseline [5] §7, neither thesis has a commercial or dedicated government owner today: **OpenStar is the only LDC company, and its path is terrestrial DT—not $D\text{-}^3\text{He}$ LDR or $p\text{-}^{11}\text{B}$ [5,14,15]**. The payoff of the boron adaptation, if closed, is an aneutronic spacecraft plant whose **astronaut shield mass is set by X-rays and space weather—not by a fusion neutron flood**.

10 Acknowledgments

The $D\text{-}^3\text{He}$ LDR framing—including the development-gap clarification in its §7—follows the systems survey maintained at [5]. Contemporary $p\text{-}^{11}\text{B}$ project and physics context is drawn from the companion survey [6]. Diligence gates and POS $\star$ follow [6] §1.7 / §12 with spacecraft adaptations noted in §8.5.

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